TABLE OF CONTENTS

1 INTRODUCTION …….05
1.1 General importance of study ……05
1.2 Aim of study ……....07
2 LITERATURE REVIEW ……..08
2.1 Factors influencing the fragmentation size of rock …….....08
2.2 Effect of Fragmentation on mining operation ……10
2.3 Optimum rock fragmentation ……..12
2.4 Significance of optimum rock fragmentation ……..13
2.5 Achievement of optimum rock fragmentation ……13
2.6 Fragmentation measurement techniques ……..13
2.7 Fragmentation analysis software ……16
3 METHODOLOGY ……20
3.1 Field study ……...20
3.2 Lab Work – Fragment size assessment by using Fragalyst 6.0 ……....23
4 RESULT & DISCUSSION ……29
5 CONCLUSION ……..32
6 FUTURE WORK ……33
7 REFERENCES ……...34

LIST OF FIGURES

Figure 2.1 Effect of degree of fragmentation on overall cost of mine …….11

Figure 2.2 Relation between the fragmentation and mining operation ……12

Figure 2.3 A large-scale muckpile image …….17

Figure 2.4 A medium-scale muckpile image ……18

Figure 2.5 Delineation of the muckpile image shown in Figure 2.4 ……18

Figure 3.1 PB area opencast mine ……...20

Figure 3.2 Bench Layout ……..21

Figure 3.3 Drilling and blasting pattern ……...21

Figure 3.4 Images taken at bench …….23

Figure 3.5 Before processing ……...24

Figure 3.6 Calibration process …….24

Figure 3.7 Pre-processing ……25

Figure 3.8 Edge detection and editing …….……26

Figure 3.9 Size distribution analysis ……26

Figure 3.10 Merged analysis …….……...27

Figure 3.11 Flow chart …….28

Figure 4.1 Blasted block size distribution results …..……..29

Figure 4.2 Insitu block size distribution results …..……...…..30

Figure 4.3 Shape Factor ……..……...…..30

Figure 4.4 Spherocity ……..……...…..30

LIST OF TABLES

Table 1 General parameters of mine ……..20

Table 2 Machinery used in mine ……22

Table 3 Pre blast survey …….22

ABSTRACT

In recent years, a vast change has taken place in the knowledge and techniques of different operations in an opencast mine. Nowadays in the market, different type of excavators and explosives are available. For a good production of a mine, the fragmentation size and cycle time of the excavator play an important role. Blast fragmentation size distributions have a direct impact on the load and haul cycle through the excavator. The fragmentation process affects not only the local productivity and unit costs of the mining department; it also influences the performance of the subsequent comminution operation, particularly when autogenously grinding is employed, and thus the throughput of the total production system.

It is in this light that the present study which is an intensive field-based programme, has been conducted in one of the most prestigious opencast coal mines. The study has been conducted to assess the effect on fragmentation under varying blast design parameters. The study was conducted on shovel benches to assess how the varying blast design parameters will affect the fragmentation size. Result obtained from this study indicates the optimum size of fragmentation and also suggests the blast design to achieve the desired fragment size so that to optimize the mine mill fragmentation system.

This study provides a relationship between blast-induced fragment size and predictions for the different fragment sizes that can be achieved on varying the blast design, which will be helpful for the mining industry to select an optimum blast design.

1 INTRODUCTION

1.1 General importance of study

In the present day economic scenario, mining of various minerals plays a very crucial role by contributing almost about 6.23% of the GDP of the country (The Hindu j. 09 feb 2011). The post independence era witnessed a massive expansion of exploration activities through various five year plans, which included the augmentation of mineral inventory as well as addition of number of mineral recourses to the existing ones.

The term fragmentation refers to post blast size distribution of the rock mass. In other words, fragmentation means economically significant size range of a definable volume. Fragmentation obtained in the blasting process influences the downstream costs like loading cost, transportation cost, processing cost etc (Singh and Yalcin, 2002, Shim et al., 2009). The fragment size should be suitable for futther handling and processing equipment in order to reduce the total production cost.

Fragmentation size and the cycle time of excavator plays a very important role in the production of mine. Blast fragmentation size distribution and thus blast design have been found to have a direct impact on the load and haul cycle through excavator dig time and bucket payload.

Previous studies have demonstrated that by reducing the excavator dig time and increasing bucket payload, significant improvements can be made in both productivity and unit cost. Simulation work reported in the literature indicates that a 20 percent improvement in dig time may result in only a three percent improvement in load and haul productivity and unit cost. At the same time, a ten percent improvement in bucket payload will directly translate to a ten percent improvement in load and haul productivity and unit cost. Based upon these findings, extensive laboratory and field work have been undertaken in the past to correlate blast fragmentation distribution to bucket payload. In contrast, the literature reports limited studies quantifying the impact of blast fragmentation on excavator dig time (Brunton et al.,2003).

The primary objective of rock blasting is fragmentation but it is extremely difficult to assess the degree of fragmentation. Sieving/screening is a direct and accurate method of evaluation of size distribution of particles or fragmentation. However, for production blasting, this method is costly, time-consuming and inconvenient. Therefore, indirect methods, such as photographic methods have been in use in blasting research (Rholl, 1987; Nie and Rustan, 1987; MacLachlan, 1989).

The digging time is only a minor fraction of the overall truck cycle time, and diggability of an excavator depends upon the muck pile shape and looseness (Hawkes et al 1995).
The diggability and the handling of ore by an excavator depend directly upon the fragmentation size of the ore body (J. Rock Mech. Min. Sci. & Geomech. Abstr, 1996).

Drilling and blasting is one of the main operations in open-pit mining, and its performance has a major impact on the overall economy of the mine. The fragmentation process affects not only the local productivity and unit costs of the mining department. It even influences the performance of the subsequent comminution steps, particularly when autogenously grinding is employed, and thus the through put of the total production system. Traditionally, mines have been divided into different departments, or cost centres, e.g. mining department and a milling department. The mining department has been responsible for the productivity and costs in the open-pit. The mining department will then try to minimise its costs for drilling and blasting by increasing spacing and hole diameters, while supplying the tonnage required by the processing plant. Since the loaders and the truck fleet are within the responsibility of the mining department, the diggability, the ease of which the ore can be dug, of the muck pile, i.e. mean fragment size, amount of boulders etc., will govern how much the specific charge may be reduced or increased. The size distribution, or fragmentation, of the ore supplied to the comminution process may have a significant impact on the throughput of the mill (Valery et al., 2001), but the relationship is not yet fully understood. It is not very likely that the optimum fragmentation that minimises the drilling and blasting costs of the mining department at the same time will maximise the throughput and minimise the production costs of the mill.

Fragmentation is the processes of breaking the solid in situ rock mass into several smaller pieces capable of being excavated or moved by material handling equipment. Breakage of rock mass is assisted by conventional drilling blasting operation which is the most important method of fragmentation in almost every quarry. There are a number of controllable as well as uncontrollable parameters that govern the fragmentation of rock. The controllable parameters can be controlled by effective blast design and use of appropriate explosive for blasting. While the uncontrollable parameters as the name suggests cannot be controlled. But certain measures have to be taken to minimize the effects of these parameters in rock blasting in order to attain an optimum rock fragmentation (Raina et al., 2002).

The efficiency of an excavator is directly related to the size distribution of muck pile. Therefore, reliable evaluation of fragmentation is a critical mining problem (S. Esen and H.A. Bolgoon, 2000).

Fragmentation of a rock dependent upon a series of factors that may include rock properties, geology, topography especially free surface conditions, explosive characteristics and finally design of the blast (Swets & Zeitlinger, 2002).

Rock fragmentation is considered the most important aspect of production blasting because of its direct effects on the costs of drilling and blasting, and it also affects the excavator cycle time (Chung et al., 2000).

Fragmentation of the rock after blasting is an important determinant of the cost associated with these two components of mine development (Kazem et al., 2005).

1.2 Aim of study

The present study was undertaken to assess the effect of fragmentation under the varying blast design parameters.

Sufficient case was taken to keep some of the parameters constant in order to critically relate the changes in mean fragment sizes.

In this light, the aim of the study is to compare the results of the predicted fragmentation size and the actual blasted fragment sizes.

2 LITERATURE REVIEW

Rock fragmentation is considered the most important aspect of production blasting because of its direct effects on the costs of drilling and blasting and on the economics of the subsequent operations of loading, hauling, and crushing.

Before starting work, it is important to know about the fragmentation parameters on which they depend.

2.1 Factors influencing the fragmentation size of rock

The fragmentation is influenced by a number of factors. Some of them we know very well, whereas others are more difficult to understand (Lung, 2009).

The conditions of the rock are of great importance, which means that the influence of changes in other factors can vary considerably from one place to another. A change of parameter in one type of rock can, for instance, give an important difference in the result, whereas a corresponding change in another place may not be noticed at all.

2.1.1 Rock condition

The physical and mechanical characteristics of the rock as well as the structure are of great importance in rock blasting (Woschnack et al., 2004). Some of the properties normally discussed are:

2.1.1.1 Tensile strength of the rock

This is normally determined by the 'Brazilian-test', which is a more relevant value than the more common compression strength. According to Cunningham (3) rock with a tensile strength >15 MPa should tend to give a rough fragmentation whereas <6 MPa is supposed to give a fine fragmentation.

2.1.1.2 Young's modulus of the rock

For the rock with the same tensile strength, a coarser fragmentation is expected at a

lower value of Young's modulus than at a higher one. According to Roke, this is most obvious when it is lower than 50 GPa.

2.1.1.3 The density and P-wave velocity of the rock

The product of the density of the rock and its P-wave velocity is called the impedance of the rock, which, according to some theories, is of great importance.

2.1.2 Specific drilling and charging

The general rule is that an increase in specific drilling and specific charge gives a finer fragmentation. This is also indicated in some way by the formulas used for the calculation of the fragmentation.

To maintain a certain fragmentation, it is therefore usual to increase the specific charge when increasing the hole diameter.

2.1.3 Drilling and firing pattern

The placement of the holes is of importance for the blasting result. This applies to the location in relation to the structure of the rock and in relation to each other.

A normally accepted rule of thumb says that for a constant area/hole, i.e., V*E = const, there will be a linear fragmentation when increasing the E/V and vice versa. By studying different tests, the best fragmentation will be found at a staggered pattern.

When a drilling and firing pattern is made, it is important to study each hole at the moment of detonation.

2.1.4 Delay times

The choice of delay times has a very big influence on the fragmentation, on one hand, that the cracks shall have enough time to develop, and on the other, that the rock must get time to move to avoid problems for the rows behind.

When choosing delay time, the risks for throw and ground vibrations must be

considered, which in certain cases can give contradictory delay times.

2.1.5 Type of explosive

In order to achieve optimal fragmentation, an explosive adapted to the actual rock conditions should be chosen. With the knowledge today, this is not easy.

Even if there are certain general rules, there are still many question marks. Among the properties of the explosive that may influence the fragmentation, there are, for instance:

- Energy content
- Gas volume
- Detonation velocity
- Density

In order to get a satisfactory picture of an explosive's rock blasting performance, it is necessary to consider its environment because of the useful energy released by the explosive is dependent on a lot of factors, like

- Properties of the explosive
- Confinement of the explosive, i.e. properties of the rock
- Hole diameter
- Degree of coupling

2.2 Effect of Fragmentation on mining operation (Mackenzie, 1967)

Mackenzie's objective was to determine the cost curves based on the mean fragmentation size. Mackenzie showed that loading, hauling, and crushing costs decreased with increasing rock fragmentation, while drilling and blasting costs increased with increasing rock fragmentation. As shown in Fig.1,

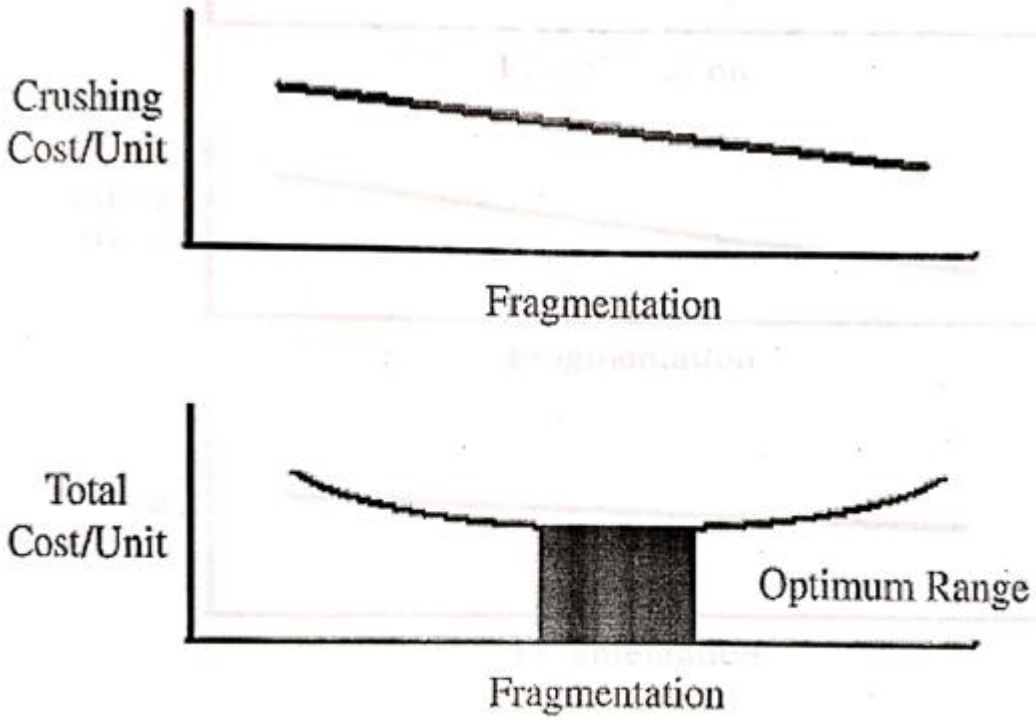


Fig. 1. Effect of degree of fragmentation on individual unit operation costs and overall cost (after Mackenzie, 1967).

Figure 2.1 Effect of degree of fragmentation on the overall cost of the mine

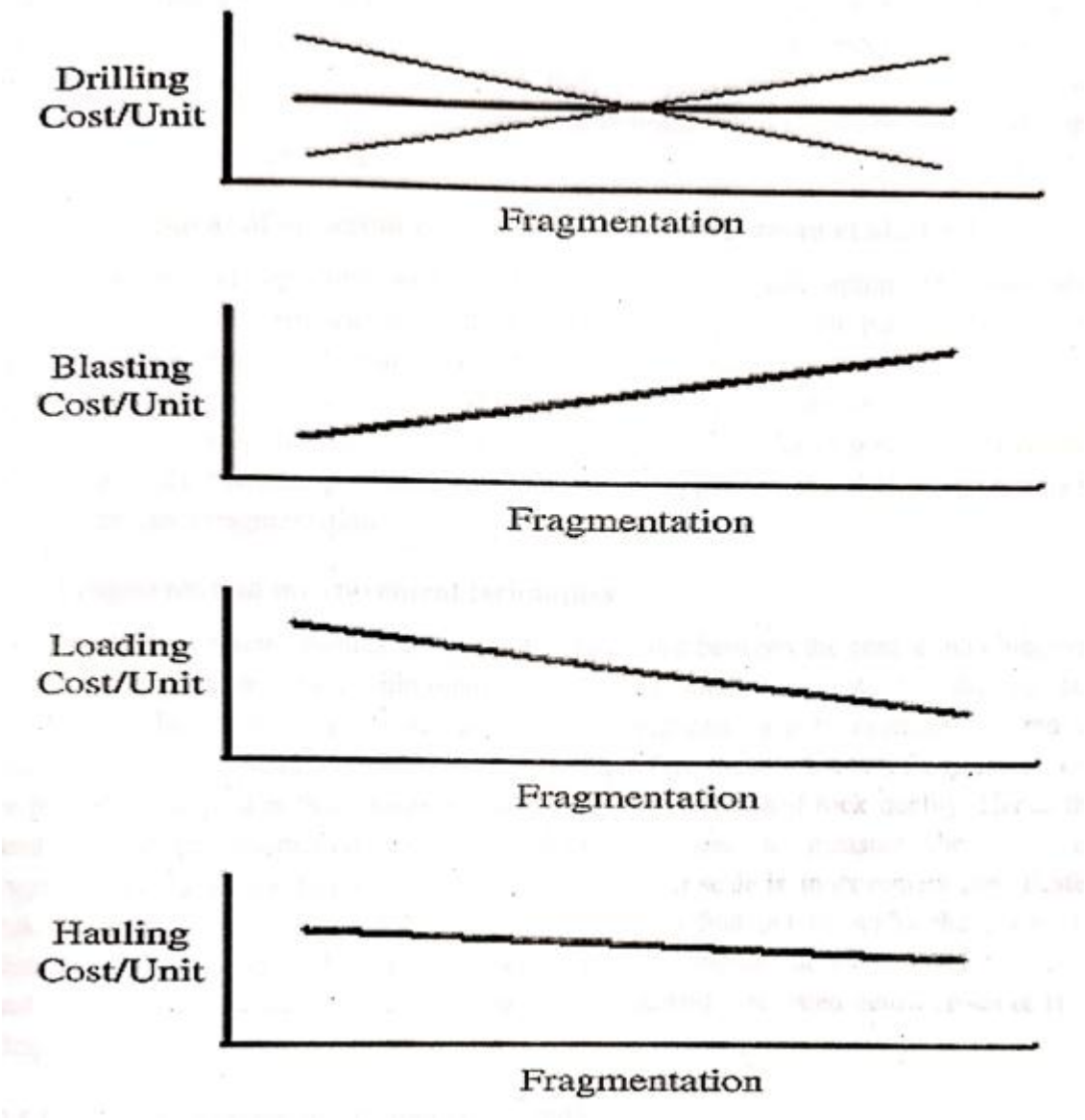


Figure 2.2 Relations between the fragmentation and mining operation

2.3 Optimum rock fragmentation (Mohanty et al., 1996)

The rock fragmentation obtained as an outcome of blasting operations is said to be optimum, when it contains maximum percentage of fragments in the desired range of size. The desired size usually means the size that is demanded and can be effectively utilized by the consumers for further operations, devoid of any processing. The desired size for different consumers is different. For example, the size of dolomite fragments required for railway tracks is comparatively smaller than the coarser ones used by the cement industry.

2.4 Significance of optimum rock fragmentation (Hartman et al., 1992)

The significance of optimum rock fragmentation is to fulfil the varying demands of different consumers for assorted sizes of rock fragments, to reduce the cost of crushing and grinding or palletisation operations, and finally uphold the economics of mining. For this the rock must be fragmented in such a way that further processing (usually termed as Milling) is not required. In other words, if the cost per ton of broken ore is greater than the price it commands when sold as the final product, then the production is not considered to be economic. Hence, the cost of milling should be minimized, and it should be ensured that the primary blast results in optimum fragmentation.

2.5 Achievement of optimum rock fragmentation (Hartman et al., 1992)

To achieve an optimum rock fragmentation, a blast with optimized controllable parameters should be designed so that the effects of the uncontrollable parameters can be minimized. The controllable parameters for optimum fragmentation can be fixed after conducting trial blasts in a mine and quantification of fragmentation. Quantification of fragmentation refers to the measurement of fragmentation in order to predict the necessary corrections in the blast design. These corrections, when applied to the blast design, result in almost acceptable fragmentation.

2.6 Fragmentation measurement techniques

Blast optimization requires a degree of compromise between the competing objectives of maximum fragmentation, minimum dilution and minimum costs for drilling and explosives. Also, mining companies and quarry operations have to examine and reduce production costs to remain competitive. But no single factor, such as cost of explosives, can be properly evaluated without measurements of fragmentation and rock quality. Hence the need to manage production costs necessitates the need to measure the post-blast fragmentation. Quantification of fragmentation on a larger scale is an extremely complicated task. Because it needs a substantial amount of time to find out manually the grain size distribution in a muckpile. Research has been carried out worldwide with different methods and tools for measurement of fragmentation. These methods are listed below (Esen & H.A. Bolgon, 1999).

2.6.1 Sieving or screening (Hinton et al.., 2006)

Sieving or screening is a direct and accurate method of evaluation of size distribution of particles or fragmentation. However, for production blasting, this method is costly, time-consuming and inconvenient. This method is feasible in case of small scale blasts. In this method the rock fragments are screened through sieves of different mesh numbers for different fragment sizes. Then the screened out fragments are grouped according to their size and the number of fragments in each size range is counted to predict the nature of the blast.

2.6.2 Oversize boulder count method (Holmberg et al., 2000)

In oversize boulder count method, manual counting of the oversize boulders in the muck pile which cannot be handled by the shovel is done. This directly gives an over-size index with respect to the total in-situ rock mass blasted. It is a very popular method of determining the post-blast fragmentation.

2.6.3 Explosive consumption in secondary blasting method (Melnikov et al., 1978)

In the Explosive consumption in the secondary blasting method, an index regarding the consumption of explosives in secondary blasting by either pop shooting or plaster shooting is determined. This index is then used for comparing the degree of fragmentation of a group of blasts.

2.6.4 Shovel loading rate method (Monjezi et al., 2009)

The shovel loading rate method assumes that the faster the mucking, the better the fragmentation. In this method, the loading rate of the shovel for a particular muck pile is taken into account. This technique may be used more accurately for a comparative account of the nature of the fragmentation of a group of blasts.

2.6.5 Bridging delays at the crusher method (Jimeno et al., 1995)

In the Bridging delays at the crusher method, the delay in bridging at the crusher mainly due to oversize boulders is observed. This attributes in determining the number of oversize boulders in the muck pile. This method is usually preferable in a small production site rather than in large scale blasting situations.

2.6.6 Visual analysis method (Maerz et al., 1996)

The Visual analysis method is a subjective assessment method. In this method the post-blast muck is viewed immediately after blasting and a subjective assessment is made. This technique is not dependable as the superficial view of the muck cannot enlighten anything about the hidden portion.

2.6.7 Photographic or manual analysis method (Sudhakar et al., 2005)

In photographic method delineating of fragments on the photographs of muck pile is carried out manually to determine the number of fragments using a graph paper. For this, 0.15m x 0.10m size photographs of the muck pile are printed. Each photograph is then placed under a transparent paper by fixing it firmly with the help of pins. All the fragments are delineated on the transparent paper. Delineation is started with large fragments because they have more effect on the results. It is tried to detect and delineate fragments as small as possible. The scale placed in the middle of the muck pile is used to convert the measured distance on the photograph to actual distance. Then, a Xerox copy of the traced paper is placed on a graph paper. The area of the reference scale on graph paper is noted down and then a scale factor (actual area of scale/graph area of scale) is determined. For every identifiable fragment, the area covered by the fragment is measured by counting the number of small blocks on the graph paper covered by that fragment. The area is then multiplied with the scale factor. For converting the area into volume, the third dimension is determined using the method of equivalent circle of area. The parameters are calculated as follows:

Equivalent diameter = (4 x Area/π), m

Spherical volume = Area x Equivalent diameter, m3

Weight of the fragment = Spherical volume x density of the rock, kg

The manual analysis of each photograph takes about one to two hours.

2.6.8 Conventional and high-speed photogrammetric method (Wallace et al., 2006)

This method is more reliable and accurate than the photographic method. It can provide three-dimensional measurements and thereby help in the calculation of fragmentation volume.

2.6.9 High-speed photography or image analysis method (Maerz et al., 1996)

Nowadays High speed photography or Digital images processing and analysis systems emerged with the advance in technology are becoming increasingly popular in fragmentation measurement. This is due to their advantages over photographic methods. Consequently several countries and organizations have developed their own image analysis systems. Some of these systems include:

- IPACS
- TUCIPS
- FRAGSCAN
- SPLIT
- Fragalyst
- WipFrag

2.7 Fragmentation analysis software

2.7.1 Description about the software

Fragalyst 6.0 is a Windows-based **Digital Image Analysis Technique (DIAT)** software designed for analyzing blasted rock fragment sizes. It operates by processing digital images of fragmented rock piles to extract data such as **mean fragment size (K50), shape factor, spherocity, and size distribution**. The software includes tools for image calibration, enhancement, segmentation, edge detection, and data interpretation.

Fragalyst supports both **in-situ block size distribution (IBSD)** and **blasted block size distribution (BBSD)** analysis, making it suitable for comparative studies on energy utilization in blasting. It provides both tabular and graphical outputs, including Rosin-Rammler and normal distribution plots, aiding decision-making in blast design optimization.

2.7.2 Accuracy of the Fragalyst 6.0

The accuracy of Fragalyst 6.0 heavily depends on the **quality of input images and correct calibration**. High-resolution images with good contrast and proper lighting (flash recommended) yield more precise results. The software's **edge detection algorithm**, enhanced with manual editing features (draw, line, polygon, rectangle

tools), allows for detailed refinement of detected fragment boundaries, thereby increasing accuracy.

Furthermore, the tool includes **fines correction** capabilities, which, when appropriately used, improve the realism of the size distribution results. For merged analyses across multiple images, density-weighted averaging further improves the representativeness of the output.

2.7.3 Image Acquisition and Scaling

Image acquisition is a critical step in fragmentation analysis. Fragalyst 6.0 requires images in common formats such as **.jpg, .jpeg, or .bmp**. To ensure precision, photographs should be taken with consistent lighting and a **scale reference (calibrator)** placed within the image.

The software offers two types of image calibration:

- **Aspect Ratio Calibration**: Accounts for camera tilt by requiring users to mark four corners of the calibrator and input its dimensions.
- **Normal Calibration**: Requires only two corners (top-left and bottom-right) and the calibrator's dimensions but doesn't correct for tilt.

Correct scaling is essential to convert pixel data into real-world measurements like fragment diameter in millimetres or centimetres.

Figure 2.3 A large scale muckpile image

Figure 2.4 A medium-scale muckpile image

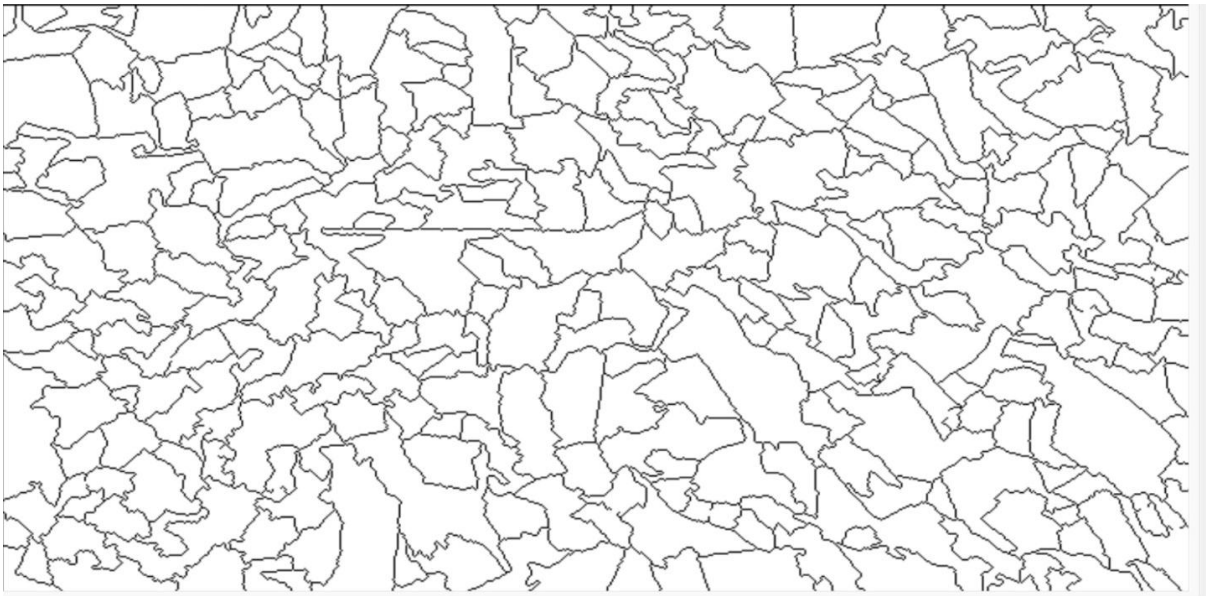


Figure 2.5 Delineation of the muckpile image shown in Figure 2.4

2.7.4 Calculation of the Size Distribution

After calibration and pre-processing, **edge detection algorithms** identify fragment boundaries. The software calculates fragment diameters, volumes, weights, shape factors, and sphericity. Users can modify or ignore detected edges through manual editing to refine results.

The size distribution is then computed using statistical models. Fragalyst supports:

- **Rosin-Rammler Distribution**: Useful for determining parameters like **K50 (mean size)** and **'n' (uniformity index)**.
- **Normal Distribution and Histogram**: For visualizing the spread and frequency of fragment sizes.

The system also computes **fines correction** for images with significant small-sized material, and **merged analysis** enables cumulative distribution over multiple blast images.

2.7.5 Presentation of the Size Distribution Results

Fragalyst 6.0 presents results in both **graphical** and **tabular formats**. Graphs like the **Rosin-Rammler curve**, **normal distribution plots**, and **cumulative histograms** are generated. Users can save these as **.jpg files** or print them directly. The software also provides the ability to **save analyzed data in .3DR format**, and to merge multiple datasets to assess overall blast performance. Results include key fragmentation metrics and are crucial for comparing predicted vs. actual fragmentation outcomes.

Additionally, users can save and export parameters related to both BBSD and IBSD, enabling a comprehensive evaluation of **blast efficiency** and **energy utilization**.

3 METHODOLOGY

3.1 Field study

3.1.1 General geology and mining details

PB Area Opencast Mine

Figure 3.1 PB area opencast mine

3.1.1.1 General parameters of mine

General parameters of mine given below in table no 1:-

Table 1 General parameters of mine

S. No.	Parameters	Details
1	Name of the mine	Gopalichak Colliery, PB Area
2	Name of Area/Company	Bharat Coking Coal Ltd.
3	Nearest Railway Station	Dhanbad, 10 km
4	Nearest National Highway	NH-18
5	Production of the mine	0.5 MT

3.1.1.2 Geological parameters

The PB Area, Gopalichak Colliery opencast mine lies within the Jharia Coalfield and is underlain by Barakar Formation rocks. Exposures are limited, and the area is structurally disturbed. Several faults traverse the leasehold, mainly concentrated in the central and southern sectors. Based on borehole data, coal seam displacements, slickensides, and sudden seam terminations, over a dozen faults have been identified. The strike of the formation ranges from N 70° E–S 70° W to N 75° W–S 75° E, with a gentle southerly dip between 5° and 12°.

3.1.1.3 Bench layout

Bench layout shown in below figure:-

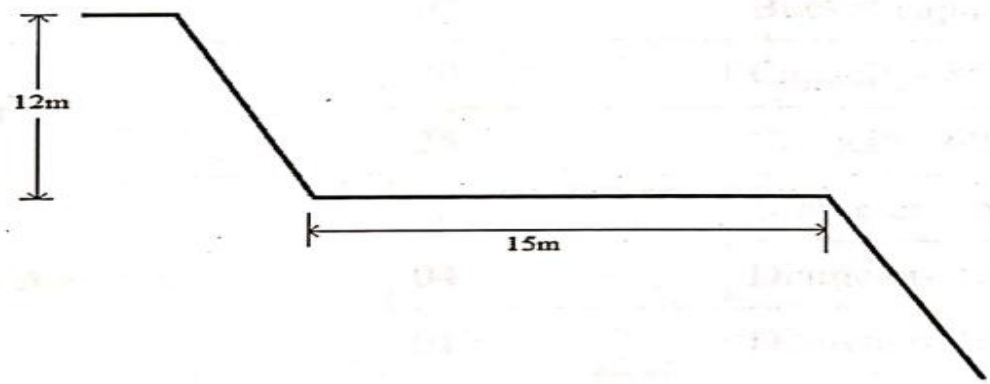


Figure 3.2 Bench Layout

3.1.1.4 Drilling and blasting

The below figure 3.4 shows the drilling and blasting pattern which is used in PB Area.

Diameter of each hole is 250mm in overburden.

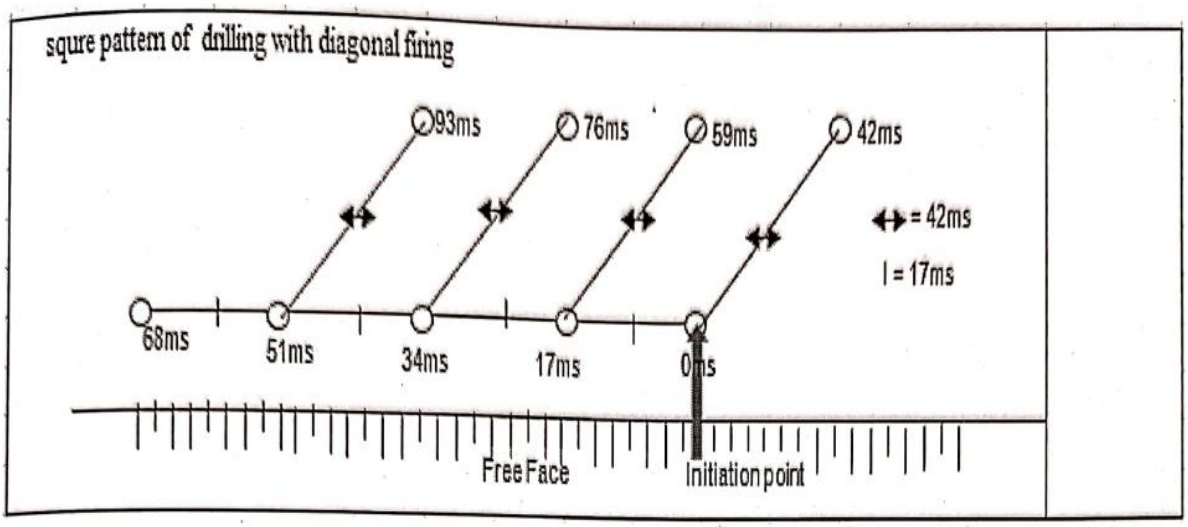


Figure 3.3 Drilling and blasting pattern

3.1.1.5 Machinery used in mine

The machineries use in PB area opencast project is given below in table no 2:-

Table 2 Machinery used in mine

S. No.	Machinery	Quantity	Capacity
1	Shovel	03	Bucket Capacity – 2.7 m3
2	Dumper	08	Capacity – 20 tonnes
3	Drilling Machine	02	Diameter – 250 mm
		02	Diameter – 160 mm

3.1.2 Pre blast survey

3.1.2.1 Drilling and blasting (overburden)

Table 3 Pre blast survey

Parameters	Bench
Bench Height	12 m
Hole Diameter	250 mm
Hole Depth	20 m to 30 m
Spacing	6 m
Burden	6 m
Drilling Pattern	Parallel
Blasting Pattern	Diagonal
Type of Explosive	Emulsion

3.1.3 Post blast survey

3.1.3.1 At bench, thirty images are taken from which six images (1-6) are shown below:-

Figure 3.4 Images taken at bench

3.2 Lab Work – Fragment size assessment by using Fragalyst 6.0

The process can be divided into several stages, which are described as:-

3.2.1 Stage 1:- Image Acquisition (Before Processing)

High-quality images of the blasted muckpile are captured using a digital camera. It is essential to ensure **uniform lighting, proper focus, and use of camera flash** for optimal image contrast. A **scale calibrator (e.g., a measuring rod or known object)** must be included in the image to facilitate accurate scaling during analysis. Images should be saved in standard formats like .jpg, .jpeg, or .bmp.

Figure 3.5 Before Processing

3.2.2 Stage 2:- Image Upload and Calibration

The captured image is uploaded into Fragalyst 6.0 using the **"Open Image"** option in the File Menu. Upon opening the image, the **calibration window** appears automatically. The image must be calibrated using one of two methods:

- **Aspect Ratio Calibration**: Corrects for camera tilt; requires marking all four corners of the calibrator and inputting its real-world dimensions (height and width).
- **Normal Calibration**: For flat images; only the top-left and bottom-right corners are marked with dimensions.

This step ensures the software correctly interprets pixel values into real-world fragment sizes.

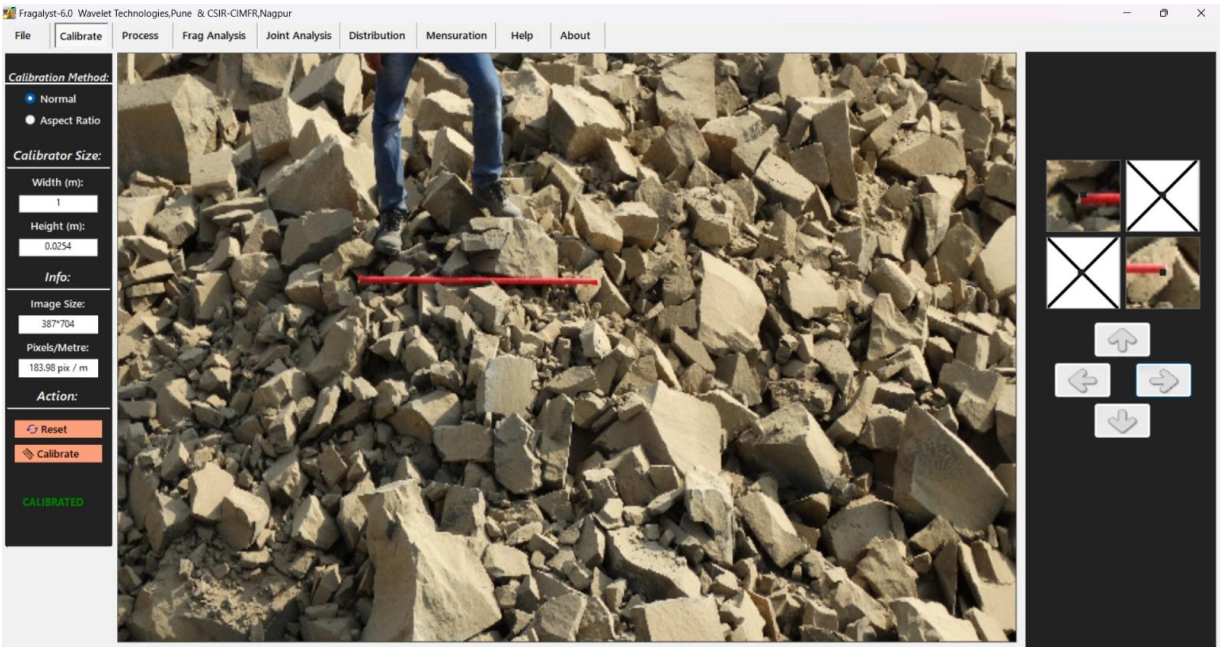


Figure 3.6 Calibration Process

3.2.3 Stage 3:- Image Pre-processing

In the **Process Image tab**, the image is enhanced to improve detection accuracy. Key pre-processing operations include:

- **Resize**: Reduces image dimensions for easier processing.
- **Brightness/Contrast Adjustment**: Performed using sliders under the Enhance menu to optimize image visibility.
- **Crop**: Allows focusing on a specific region of interest.
- **Negate/B&W**: Converts image to grayscale for better fragment boundary recognition.

Once edits are applied using the **Apply** button, the image is ready for analysis.

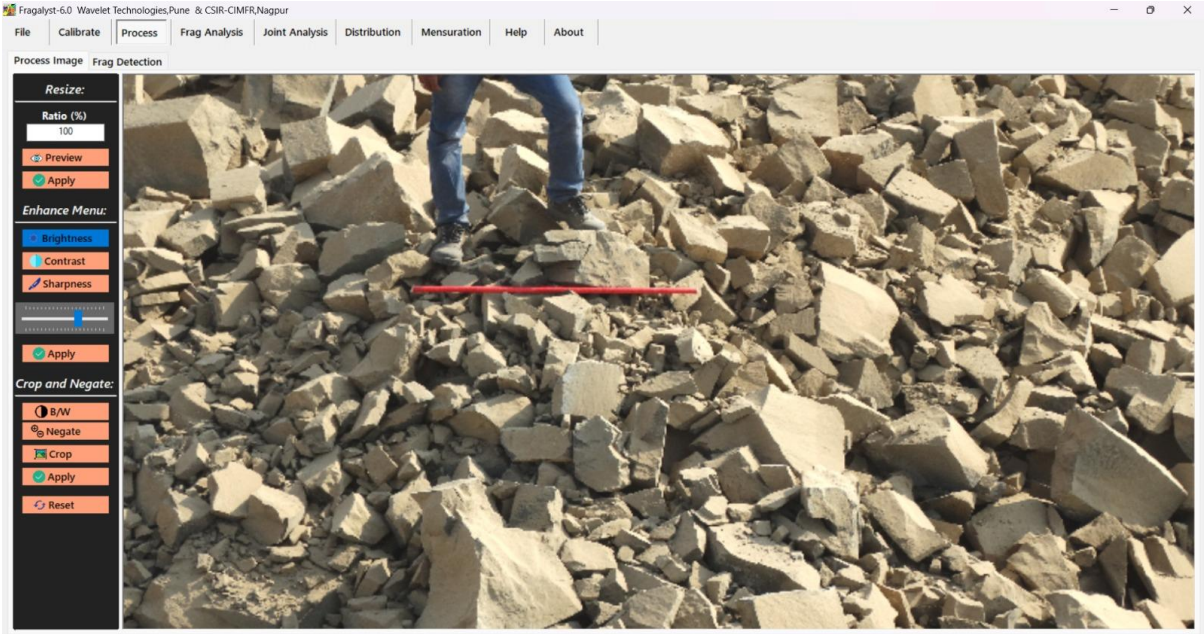


Figure 3.7 Pre-processing

3.2.4 Stage 4:- Fragment Detection and Boundary Editing

In the **Frag Detection tab**, the software performs **edge detection** to identify the boundaries of individual fragments. The user can adjust the **threshold values** to fine-tune the detection.

Manual editing tools such as:

- **Draw**, **Line**, **Free Line**, **Rectangle**, and **Ignore**

are available for correcting the automatically detected fragment edges. The **boundary color** can also be customized for better visibility.

After editing, the **Finish** button is clicked to finalize the segmentation. This action also saves the data in .3DR format automatically.

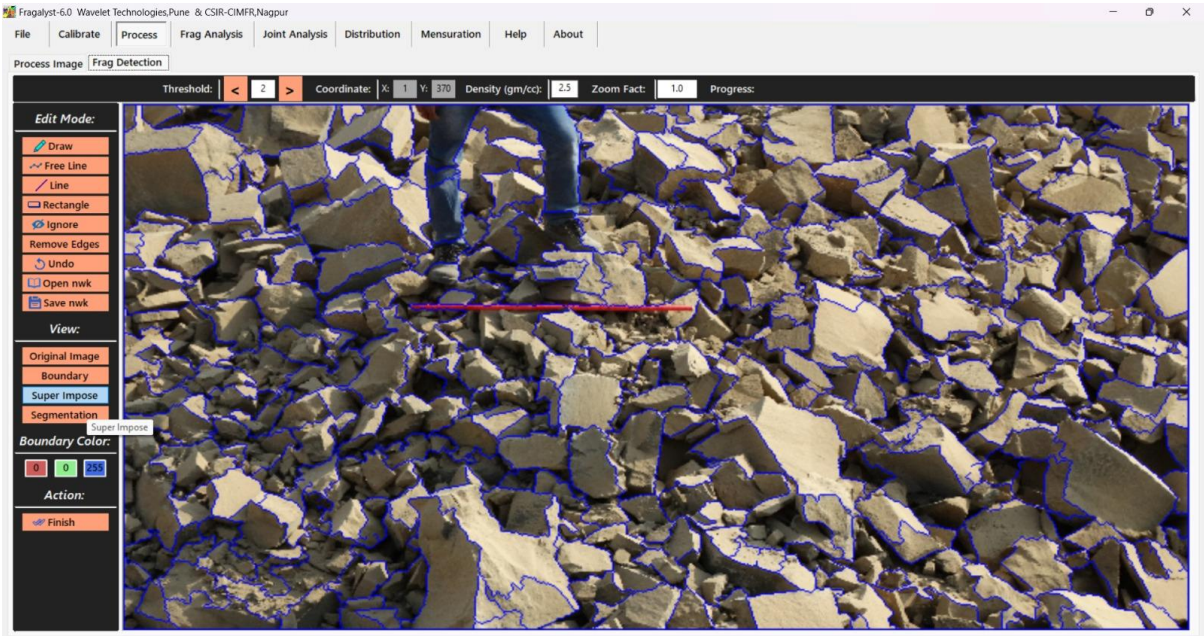


Figure 3.8 Edge detection and editing

3.2.5 Stage 5:- Fragment Size Distribution Analysis

Upon completion of fragment detection, the software shifts to the **Frag Analysis tab**, where it calculates and displays:

- **Total number of fragments detected**
- **Rosin-Rammler Curve** (with K50 and uniformity index 'n')
- **Normal Distribution Plot**
- **Cumulative Histogram**

The user can apply **Fines Correction** if the fragmentation includes a significant number of fines. The analysis graphs and data can be saved or printed for documentation.

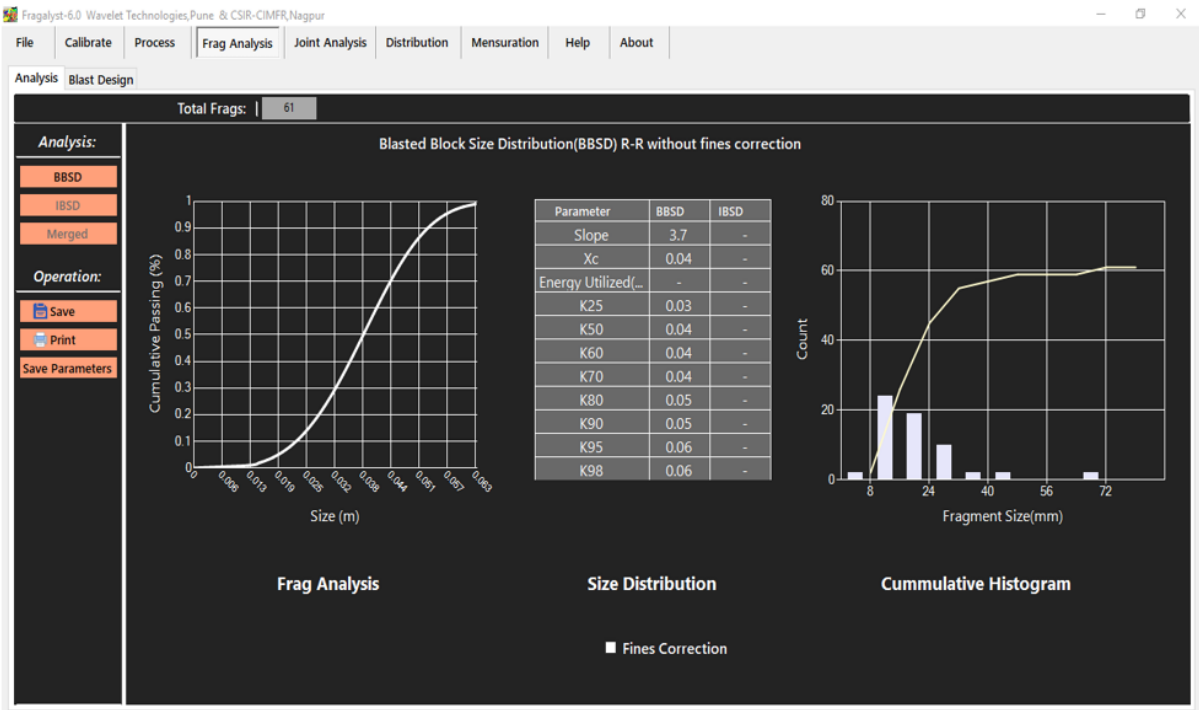


Figure 3.9 Size distribution analysis

3.2.6 Stage 6:- Merged Analysis (If Applicable)

When multiple images are analyzed for a single blast event, their `.3DR` files can be merged to assess overall blast performance. This is particularly useful for obtaining **representative average fragment sizes** across the entire muckpile. The merged analysis automatically weights each image based on its input **density** and provides a combined distribution.

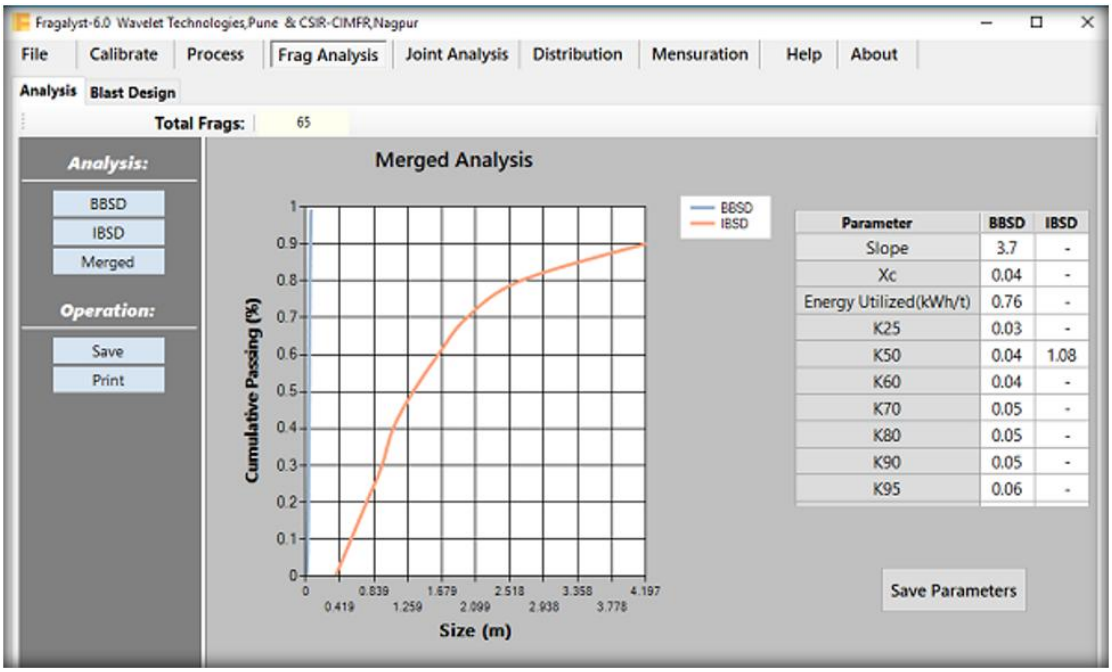


Figure 3.10 Merged analysis

3.2.7 Stage 7:- Saving and Reporting

All graphical and tabular results from the analysis can be:

- **Saved as image files** (.jpg)
- **Saved as text reports** for parameters
- **Printed** directly from the software

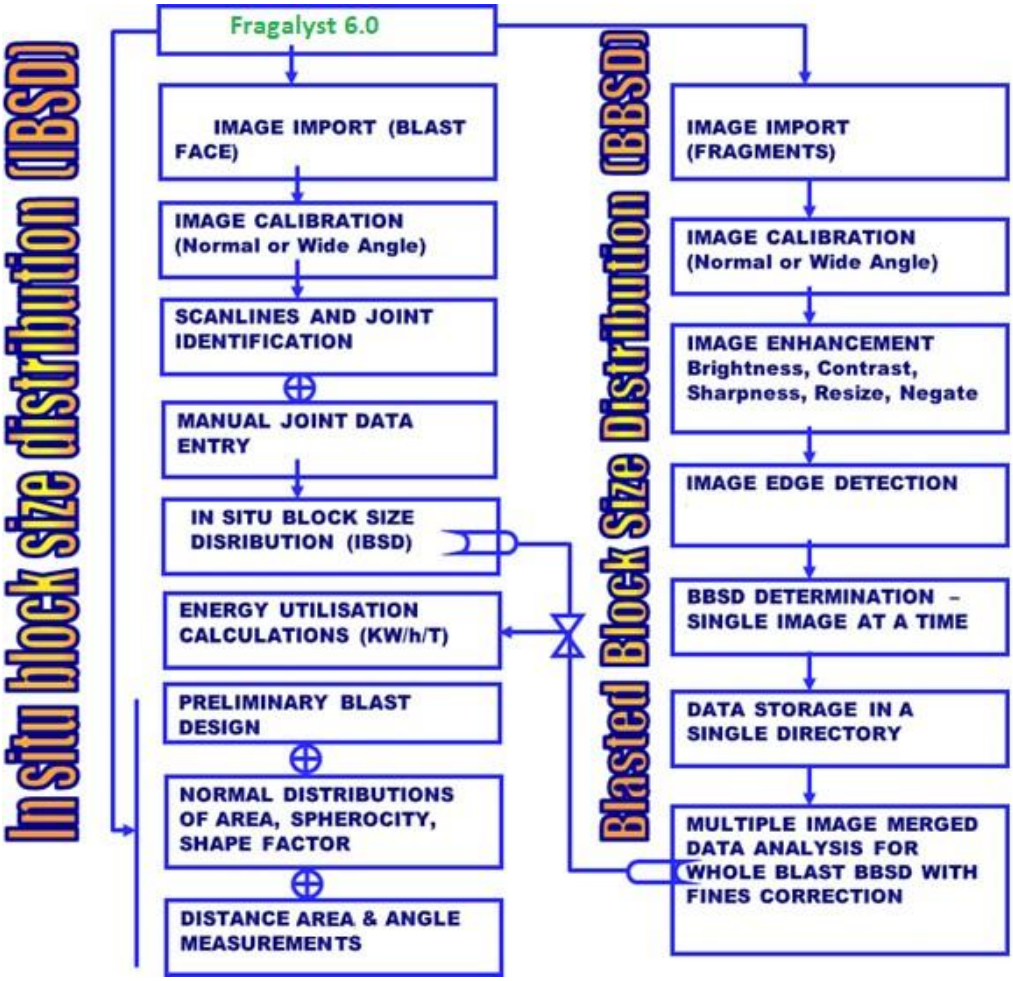
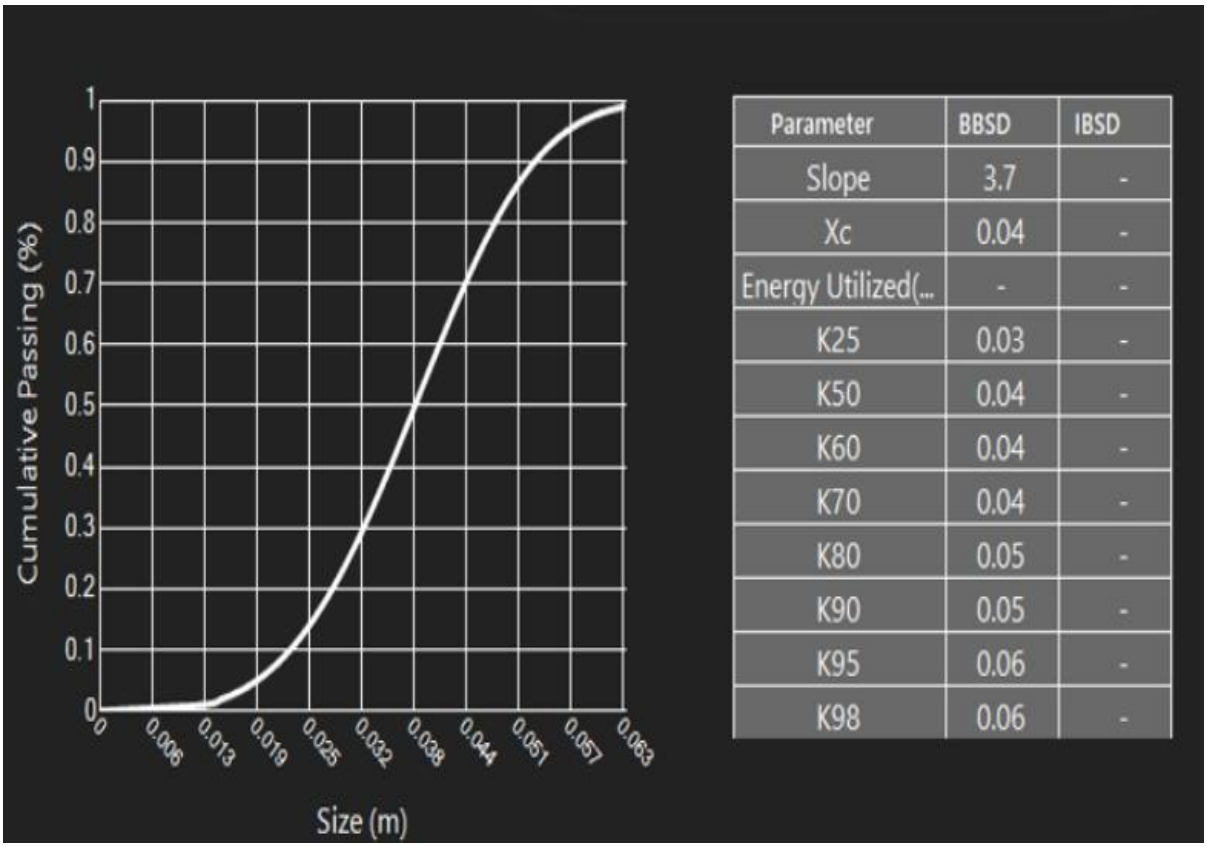


Figure 3.11 Flow chart of processes of fragalyst 6.0

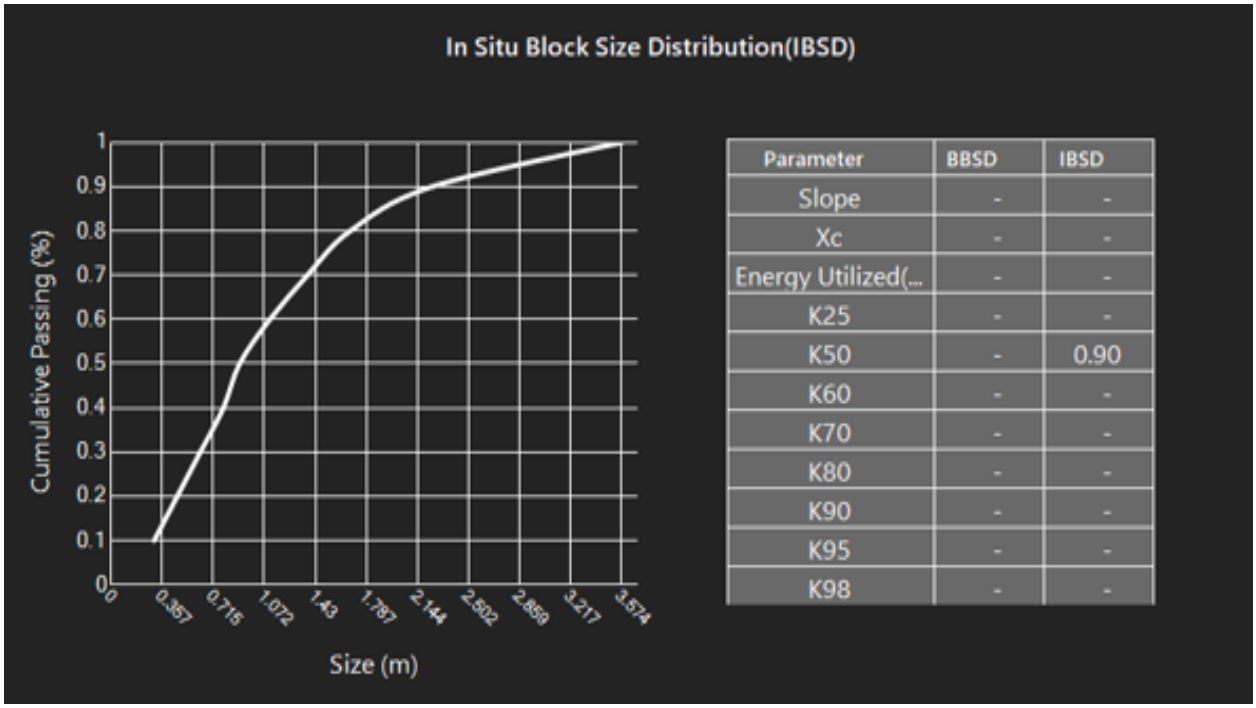
4 RESULT & DISCUSSION

The fragmentation analysis was carried out using Fragalyst 6.0 software, which employs digital image analysis techniques to estimate the fragment size distribution of blasted rock piles. The software allowed for image calibration, boundary detection, and fragment size computation under varying blast design parameters. The results obtained from this analysis are discussed below.



Parameter	BBSD	IBSD
Slope	3.7	-
Xc	0.04	-
Energy Utilized(...	-	-
K25	0.03	-
K50	0.04	-
K60	0.04	-
K70	0.04	-
K80	0.05	-
K90	0.05	-
K95	0.06	-
K98	0.06	-

Figure 4.1 Blasted Block Size Distribution results



Parameter	BBSD	IBSD
Slope	-	-
Xc	-	-
Energy Utilized(...	-	-
K25	-	-
K50	-	0.90
K60	-	-
K70	-	-
K80	-	-
K90	-	-
K95	-	-
K98	-	-

Figure 4.2 In-situ Block Size Distribution results

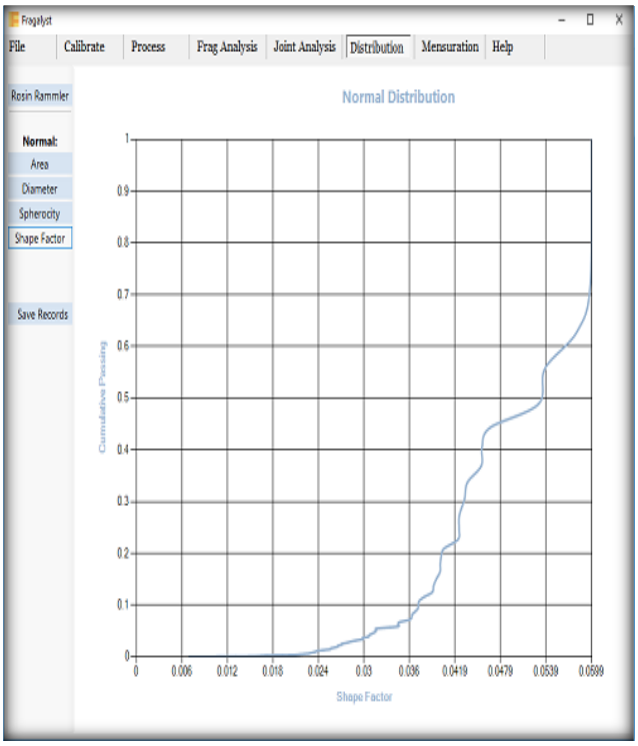


Figure 4.3 Shape Factor

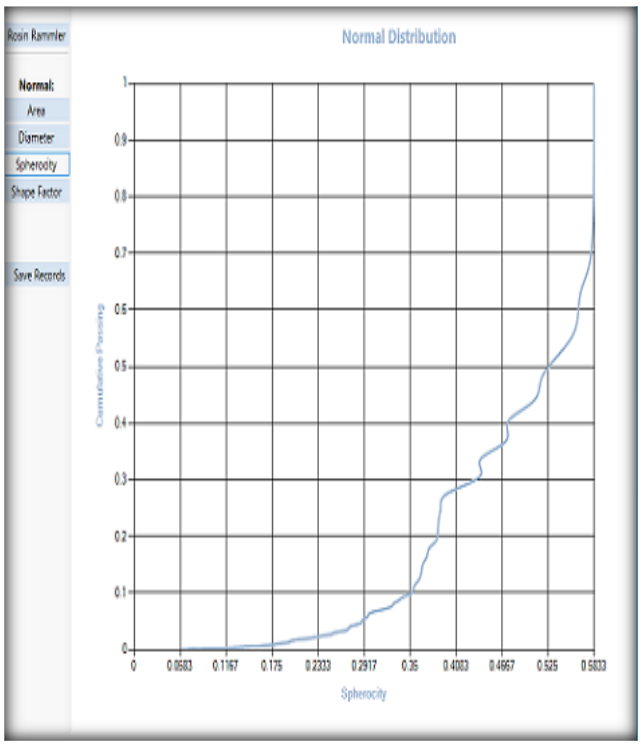


Figure 4.4 Sphericity

- Blast parameters like burden, spacing, and charge directly influenced fragmentation. Optimized designs showed lower K50 values and better uniformity, while poor designs resulted in coarse and uneven fragmentation.
- Accurate image calibration using aspect ratio correction ensured reliable measurements. Manual editing tools helped refine fragment boundaries, especially in complex or unclear images.
- Fines correction improved results where small fragments were dominant. Merged analysis of multiple images gave a comprehensive view of blast performance, especially when density-based weighting was applied.
- The built-in prediction tool estimated expected fragmentation based on input parameters. Predictions closely matched actual results, supporting its usefulness in pre-blast planning.

5 CONCLUSION

From the above study it may be concluded that:-

- **Digital image analysis is a reliable and effective method** for evaluating blast fragmentation in surface mining.
- **Blast design parameters significantly influence fragmentation**, with optimized designs leading to finer and more uniform fragment sizes.
- **Fragalyst 6.0 provides accurate fragment size distribution curves**, especially when supported by proper calibration and manual editing of boundaries.
- The **blast prediction module** within the software aids in pre-blast planning and has good correlation with actual fragmentation results.
- **Merged image analysis and fines correction** improve the reliability of overall fragmentation assessment when multiple data sets are involved.
- Proper field practices such as placing a calibrator and maintaining good image quality are essential for ensuring analysis accuracy.

6 FUTURE WORK

1. **Field Validation:**
   Future studies can compare Fragalyst results with physical sieve analysis or direct field measurements for improved accuracy.
2. **Use of Advanced Imaging:**
   Incorporating drone-based or 3D photogrammetry can enhance image quality and coverage for more reliable fragmentation assessment.
3. **Automation and AI:**
   Applying machine learning for automatic boundary detection can reduce manual editing and improve processing efficiency.
4. **Linking with Operational Performance:**
   Analyzing the impact of fragmentation on equipment productivity and processing costs can help optimize overall mine performance.

7 REFERENCES

1. Chung et al. (2000): Fragmentation prediction using improved engineering formula. FRAGABLAST - Int. J. Blast. Fragment. 4, 198-207.
2. Raina et al. (2002): Fragalyst — an indigenous digital image analysis system for grain size measurement in mines. J. Geol. Soc. India 59, 561—569,
3. Maerz et al. (1996). Wipware, Inc., Canada
4. Rholl, 1987; Nie and Rustan, 1987; MacLachlan, 1989.
5. Kemeny et al. (1993): "Analysis of Rock Fragmentation Using Digital Image Processing", Journal of Geotechnical Engineering, Vol. 119, No. 7, pp.1144-1160.
6. Hawkes et al. (1995): Drilling and Blasting of Rocks.
7. Higgins et al. (1999): "Integrated Software Tools and Methodology for Optimization of Blast Fragmentation", Proceedings of the Twenty-Fifth Annual Conference on Explosives and Blasting Technique, Nashville, Tennessee, USA, Volume II, pp. 355-368.
8. Kemeny et al. (1999): "New Advances in Digital Image Analysis Software to Quantify the Size Distribution of Fragmented Rock", MINNBLAST 99, pp. 27-43.
9. Franklin et al. (1996): "Evolution of Measuring Systems: A review", Proceedings of the Fragblast-5 Workshop on Measurement of Blast Fragmentation, pp. 47-52.
10. Kemeny et al. (1994): "A Practical Technique for Determining the Size Distribution of Blasted Benches, Waste Dumps, and Heap-Leach Sites", Mining Engineering, Vol. 46, No.11, pp.1281-1284.
11. Esen and Bolgoon, (2000): "Comparing systems - Validation of Fragscan, Fragalyst, and Split", Proceedings of the Fragblast-5 Workshop on Measurement of Blast Fragmentation, pp. 151-155..
12. Franklin et al. (1987): Photographic Measurements of Jointing and Fragmentation. Proc. 2nd Int. Symp. Field Measurements in Geomechanics, Kobe, Japan, v 1 ,pp. 1-11.
13. Kemeny (1994): Practical technique for determining the size distribution of blasted benches, waste dump and heap leach sites. Min. Engg, pp 1281—1284.
14. Maerz et al. (1996): Fragalyst image based granulometry system. In: Franklin, J. A., Katsabanis, P. D. (eds.), Measurement of blast fragmentation. Balkema, Rotterdam, 1996, pp 91—98.
15. Wallace et al., (2006):FRAGSCAN: A tool to measure fragmentation of blasted rock.ln: Franklin, J. A., Katsabanis, P. D. (eds.), Measurement of blast fragmentation. Balkema, Rotterdam, pp 73-78.
16. Woschnack et al. (2004): Rock Mechanics and Rock Engineering.
17. Sudhakar et al., (2005): Photographic or manual analysis method.
18. Mackenzie, 1967: effect of fragmentation on mining operation.